

GABRIEL REYNA ALVARADO

Data Analyst | Machine Learning

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PROFILE

Computer Science student with hands-on experience across the full data project lifecycle: dataset preparation and cleaning, training and evaluation of classification models, and deployment of results in dashboards for non-technical users (Tableau and Streamlit). I work with explicit methodological rigour, including data leakage detection, metric selection driven by the cost of each error type, and documented limitations, on a Python, SQL and scikit-learn/XGBoost stack.

DATA AND MACHINE LEARNING PROJECTS

ALDIMI Predict — Predictive system for a paediatric oncology shelter

Python, XGBoost, Scikit-learn, SHAP, SQLite, Streamlit | UPC, 2026 | Team project: modelling and deployment

- Trained and evaluated classification models (XGBoost, Random Forest) to anticipate 7- and 14-day stockouts in the inventory of a shelter for paediatric oncology patients, tuning hyperparameters with RandomizedSearchCV over 5-fold StratifiedKFold.
- Built a multiclass clinical risk classifier over 34 nutritional, socioeconomic and treatment-adherence variables, reaching a macro F1 of 0.75 and AUC of 0.87 on a synthetic dataset generated to protect the confidentiality of real patients.
- Interpreted predictions using SHAP values to surface the dominant risk factors, a prerequisite for shelter staff to audit a model that informs care decisions.
- Deployed the system as an interactive Streamlit dashboard on a unified SQLite database, delivered to the organisation together with a user manual and a data dictionary.

Smart Kitchen Intelligence — Food waste analytics

Python, Polars, Pandas, Scikit-learn, Tableau | UPC, 2026 | Team of 3: data preparation, modelling and visualisation

- Implemented the cleaning pipeline over 25,819 inventory movements, applying 6 verifiable quality rules (outliers, category harmonisation, referential integrity, structural nulls, deduplication), with cross-implementation in Polars and Pandas for independent validation.
- Trained waste classification models (Random Forest against Logistic Regression), detecting and removing a data leak that inflated metrics to $F1 = 1.0$, and selected the final model on F1 and cross-validation stability rather than accuracy, since a trivial classifier already reached 64.5%.
- Designed a 10-sheet Tableau dashboard with 4 KPIs, a global household filter and a prescriptive panel, which identified that 97% of the economic loss concentrated in a single storage location, with a waste rate 9.1 times higher than the most efficient one.
- Validated the representativeness of the simulated dataset against the UNEP Food Waste Index Report 2024 benchmark, and documented the structural redundancy between location and product category as a methodological limitation.

Food waste recommender engine

Python, Scikit-learn, ALS, NetworkX, Streamlit | UPC, 2026 | Team project: modelling and deployment

- Trained a three-signal hybrid recommender (TF-IDF content similarity, implicit ALS collaborative filtering, and expiry-proximity urgency), evaluated under a blind masked basket completion protocol and achieving 100% catalogue coverage against the popularity baseline.
- Segmented consumption patterns with DBSCAN after benchmarking against K-Means and GMM on the PCA-reduced space, reaching a Silhouette Score of 0.65 with under 0.3% noise.
- Empirically ruled out adding PageRank centrality to the ensemble through an ablation over the weight grid (optimal weight of zero), documented the negative result in the technical report, and deployed the system as a Streamlit application.

Food condition classification with computer vision

Python, CNN, OpenCV | UPC, 2025

- Trained a convolutional neural network for multiclass classification of fruit preservation state, with an OpenCV preprocessing pipeline (normalisation and noise reduction) to improve training efficiency.

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Cirion Next Generation

Lima, Peru | January 2024 – March 2024

- Processed and cleaned datasets of over 10,000 records using SQL and Python (Pandas), implementing consistency checks that eliminated manual rework in engineering reports.
- Automated the generation of recurring visualisations with Matplotlib and Seaborn, freeing roughly 20 hours of manual work per month for the team.
- Refactored SQL queries alongside the data engineering team, reducing execution times for business-critical reports.

Simulation Developer — UPC GameLab

Lima, Peru | March 2024 – July 2025

- Modelled real-time simulation systems using object-oriented programming and optimised data structures, reducing processing overhead by 15% against the baseline implementation.
- Established the version control workflow with Git for a repository of over 500 binary assets, eliminating integration conflicts during team work.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, R, JavaScript

Data and machine learning: Pandas, Polars, NumPy, Scikit-learn, XGBoost, SHAP, OpenCV

Techniques: Supervised classification, clustering (DBSCAN, K-Means, GMM), PCA, t-SNE, collaborative filtering (ALS), TF-IDF, cross-validation

Visualisation: Tableau, Streamlit, Matplotlib, Seaborn, Plotly

Databases: SQL Server, MySQL, SQLite, MongoDB, relational modelling

Tools: Git, GitHub, Docker, pytest, Excel

Methodologies: CRISP-DM, Scrum

EDUCATION

B.Sc. in Computer Science — Universidad Peruana de Ciencias Aplicadas (UPC)

Lima, Peru | Expected graduation: 2026 Relevant coursework: Machine Learning, Data Visualisation, Database Management, Statistical Analysis.

Computer Vision & AI Programme — University of Pittsburgh

Remote | 2022 – 2025 International collaboration programme.

LANGUAGES, CERTIFICATIONS AND ACTIVITIES

Languages: Spanish (native), English (advanced), French (intermediate)

Certifications: Scrum Fundamentals Certified (SFC) | LinkedIn Certifications

Activities: Hackathon, PMI Lima Peru Chapter, PMI LATAM Conference 2026. Inter-university innovation competition on SDG 11 (Sustainable Cities and Communities); member of a team of 6 that developed and defended a proposal before an industry jury.